Import Data using Transform Maps (Spreadsheet)

Phase-7 -- Project Documentation Phase

Date	29.09.2026
Team Id	SWTID-2026-2101
Project Name	Import Data using Transform Maps (Spreadsheet)
Project Guide Name	Mrs. Rajashyamala, M.Sc., M.Phil., B.Ed., TET

Team Members:

Arshitha A -Team Leader	24721ER002
Cathrin Mary A	24721ER003
Santhiya S	24721ER010
Suvina Mary A	24721ER011

PROJECT DOCUMENTATION OVERVIEW

The Project Documentation Phase captures the completed ServiceNow implementation that imports employee data from a spreadsheet into a controlled target table. Project documentation is the authoritative record of configuration decisions, operational procedures, data flow, testing artifacts, and final results. Its purpose is to preserve implementation details so instructors, administrators, and future maintainers can reproduce, audit, or extend the solution with confidence.

This document documents the entire employee spreadsheet import process implemented in ServiceNow. It records the configuration of the Import Set (Employee Import staging table), the Transform Map (Sample Spreadsheet Import), the Coalesce behavior using Employee ID, the Employee Test target table, and the reporting/dashboard artifacts. The documented flow is:

Spreadsheet Data → Employee Import → Transform Map → Coalesce → Employee Test → Reports → Dashboard

Clear implementation records improve maintainability by explaining how data is prepared, staged, transformed, and validated; they also capture testing evidence, transform history results, and the final analytics that demonstrate successful import behavior.

PROJECT OBJECTIVES

The project objectives describe specific, implemented goals realized during the ServiceNow import exercise. Each objective below is aligned to the actual configuration delivered.

1. Import employee data from a spreadsheet into ServiceNow: Implemented by loading an Excel file into an Import Set.
2. Use an Import Set table as a staging area: Employee Import (u_employee_import) serves as the staging table for raw spreadsheet rows.
3. Create a custom Employee Test target table: Employee Test (u_employee_test) was created to receive transformed records.
4. Configure a Transform Map between source and target tables: Transform Map “Sample Spreadsheet Import” maps Employee Import to Employee Test.
5. Map spreadsheet fields to target fields: Field-to-field mappings were created for Employee ID, Employee Name, Email, Department, and Location.
6. Transform employee records into the target table: The transform execution iterates import set rows to insert or update target records.
7. Use Coalesce to identify existing employee records: Employee ID is set as the Coalesce field to match existing records.
8. Insert new employee records: New Employee IDs in the spreadsheet create new records in Employee Test.
9. Update existing employee records: When Employee ID already exists, changed fields are updated on the target record.
10. Prevent duplicate records during repeated imports: Coalesce prevents creating duplicates when the

same spreadsheet is imported again.

11. Create reports based on employee data: Three reports were produced (by department, by location, and a list).
12. Create a dashboard containing the reports: Employee Analytics Dashboards contains the three reports for centralized review.
13. Provide centralized visibility of employee information: The dashboard aggregates the reports for stakeholders.

PROJECT DOCUMENTATION SCOPE

This documentation covers all implemented components and procedures required to reproduce and validate the completed import process. The scope includes preparation through final validation and excludes any unauthorized or unimplemented features.

- Spreadsheet preparation: detailed header and row expectations.
- Employee Test target table: structure, fields, and validation steps.
- Employee Import staging table: usage and lifecycle.
- Transform Map: configuration, mapping assist, and execution steps.
- Field mapping: exact source-to-target field definitions.
- Transform execution: step-by-step procedure and Transform History checks.
- Coalesce configuration: Employee ID usage and repeated-import behavior.
- New record insertion and existing record updates: observed behavior and examples.
- Repeated import behavior and Transform History verification.
- Reports and Dashboard: report definitions and dashboard assembly.
- Final validation and project conclusion: evidence of successful import and analytics availability.
- ServiceNow environment and navigation used during implementation.
- Custom Employee Test table creation and field configuration.
- Spreadsheet data loading and staging verification.
- Source-table and target-table relationship documentation.
- Transform execution status and success verification.
- Employee record verification after transformation.
- Import result tracking through inserted, updated, and ignored record counts.
- Verification of changed employee information after updates.
- Report configuration, dashboard organization, and access documentation.

SYSTEM AND DATA FLOW DOCUMENTATION

This section details every stage of the implemented data flow from the spreadsheet to analytics. The following table summarizes components, purpose, and results for each stage.

Stage	Component	Purpose	Result
4.1 Spreadsheet Data Preparation	Local Excel file (Sheet 1)	Prepare structured employee rows with header row defining columns	Excel file containing 4 test rows; headers: Employee ID, Employee Name, Email, Department, Location
4.2 Employee Import Staging	Employee Import (u_employee_import)	Receive uploaded spreadsheet rows as Import Set records	4 rows staged for transformation
4.3 Transform Map	Sample Spreadsheet Import	Map source fields to Employee Test target fields and facilitate transform	Configured with Auto Map Matching Fields and Mapping Assist
4.4 Field Mapping	Source → Target mappings	Translate import column values into corresponding target fields	Direct mappings for Employee ID, Employee Name, Email, Department, Location
4.5 Coalesce	Transform Map coalesce setting	Use Employee ID to identify existing target records and avoid duplicates	Existing records matched and updated; new records inserted
4.6 Transformation	Transform execution	Process staged rows into target table using mapping and coalesce logic	2 inserts and 2 updates on first run; repeated run generated 0 inserts/0 updates

4.7 Target Table	Employee Test (u_employee_test)	Persist transformed employee records for reporting	Populated with the transformed data reflecting inserts and updates
4.8 Reports	Three ServiceNow reports	Provide aggregated and list views of Employee Test data	Reports created: Employees by Department, Employees by Location, Employee List Report
4.9 Dashboard	Employee Analytics Dashboards	Combine three reports into a single view for stakeholders	Dashboard contains all three reports for centralized review

SPREADSHEET DOCUMENTATION

The spreadsheet is the primary data source used in the implementation. It is prepared as a single-sheet Excel file (Sheet 1) with a header row and structured rows for each employee. The documented test scenario used four rows to validate insert and update behavior.

Spreadsheet details:

- Header Row: Row 1 contains column labels: Employee ID, Employee Name, Email, Department, Location.
- Sheet Number: Sheet 1 (used during import).
- Employee records: 4 rows uploaded in the test import.
- New employee records: 2 rows with previously unseen Employee IDs.
- Existing employee records: 2 rows matching existing Employee IDs but with modified fields.
- Modified information examples: Raviteja's email changed from test10@gmail.com to test18@gmail.com;

SB-OOO4 name changed from Ajay kumar to Ajay.

Field documentation table:

Field	Description	Usage
Employee ID	Unique identifier for an employee (string)	Used as Coalesce key to match existing records
Employee Name	Full name of employee (string)	Mapped to Employee Test name field
Email	Primary email address (string)	Mapped to target email; used for contact and validation
Department	Organizational department (string)	Used for grouping in reports (Employees by Department)
Location	Work location or office (string)	Used for grouping in reports (Employees by Location)

EMPLOYEE TEST TABLE DOCUMENTATION

The Employee Test table is the custom target table created to store transformed employee records. It is the canonical repository for the imported employee data used in reporting and verification.

Target table metadata:

Label	Employee Test
Name	u_employee_test

Field documentation and purpose:

Field Name	Type	Purpose
Employee ID	String	Primary identifying attribute; used for Coalesce matching and record lookup
Employee Name	String	Human-readable employee name displayed in lists and reports
Email	String	Contact attribute for each employee
Department	String	Organizational grouping used in aggregated reporting
Location	String	Geographic or site grouping used in aggregated reporting

Why the target table is required: It isolates transformed and cleansed data from raw staged data, enabling reporting and validation without altering source rows. Validation occurs by reviewing Employee Test records after transformation and cross-checking key fields against the spreadsheet source.

EMPLOYEE IMPORT TABLE DOCUMENTATION

The Employee Import table functions as the Import Set staging table that temporarily holds each spreadsheet row as an import set record. This staging approach preserves raw input and allows transformation logic to be applied in a controlled manner.

Label	Employee Import
Name	u_employee_import

Component-level description:

Component	Configuration	Purpose
Import Set Table	u_employee_import	Store raw rows uploaded from the spreadsheet for subsequent transforms
Spreadsheet Loader	File import (Excel) into Import Set	Populate staging table using Sheet 1 and Header Row 1
Temporary Data	Staged import records	Allow inspection and correction prior to transform execution
Relation to Target	Source for Transform Map	Serves as the translation source feeding Employee Test

The staging table's role is to decouple raw input from the target table so that the Transform Map and Coalesce logic can reliably create or update Employee Test records while preserving an audit trail in Transform History.

TRANSFORM MAP DOCUMENTATION

The Transform Map centralizes mapping rules and transform options that move data from Employee Import to Employee Test. It was implemented with settings that support reliable field mapping and coalescing.

Configuration Item	Value
Transform Map Name	Sample Spreadsheet Import
Source Table	u_employee_import (Employee Import)
Target Table	u_employee_test (Employee Test)
Auto Map Matching Fields	Enabled
Mapping Assist	Used to confirm and refine field mappings
Coalesce Field	Employee ID
Execution	Manual run of transform against selected import set

Purpose and behavior: The Transform Map converts import set columns into target fields, applies the Coalesce rule (Employee ID) to determine insert vs update, and writes results to Employee Test. Mapping Assist and Auto Map simplify mapping creation while explicit map entries ensure field-level accuracy.

FIELD MAPPING DOCUMENTATION

Accurate mapping ensures that values from the spreadsheet populate the correct target fields in Employee Test. Mapping entries were created and verified using Mapping Assist alongside Auto Map matching.

Source Field	Target Field	Purpose
Employee ID	Employee ID	Unique identifier used for Coalesce matching and target record identity
Employee Name	Employee Name	Populate the target name field for display and reporting
Email	Email	Contact value retained in target for communications and verification
Department	Department	Used for aggregated reporting and organizational grouping
Location	Location	Used for aggregated reporting and site-based grouping

Correct field mapping supports accurate transformation by ensuring each column's semantics are preserved when creating or updating Employee Test records.

COALESCE DOCUMENTATION

Coalesce prevents duplicate target records by using a unique identifying field to detect matches between staged rows and existing target records. In this implementation Employee ID is the coalesce key, which ensures deterministic insert/update behavior.

Why duplicates occur: Repeated imports or multiple rows with the same identifier will create duplicates unless a coalesce rule is set. Purpose of Coalesce: match staged rows to existing records so transforms perform updates instead of blind inserts.

Scenario	Uploaded	Inserted	Updated	Ignored	Result
First import (test spreadsheet)	4	2	2	0	Demonstrates inserts for new IDs and updates for existing IDs
Repeated import (same spreadsheet)	4	0	0	4	All rows ignored due to Coalesce matching exact existing records

How recognition works: During transform execution the engine compares the Employee ID value from the import row to existing Employee Test records. If a match exists, the record is updated; if no match exists, a new record is inserted. Repeated import shows Coalesce prevents re-insertion or re-updating when no field changes occur.